

Exercise 1.

a) $M(A, k)$:= the k -th element of the sorted version of A

1. Split A randomly in $\frac{n}{7}$ groups of 7 elements.

2. Determine the median of each group of 7 elements.

3. Determine recursively the median of these $\frac{n}{7}$ medians, say m .

4. Count the number of elements in A that are $\leq m$, say l .

- If $l = k$, m is the output

- If $l > k$, then m is larger than the number we are looking for so we throw out all the numbers that are larger $M(A \setminus A_{\text{high}}, k)$

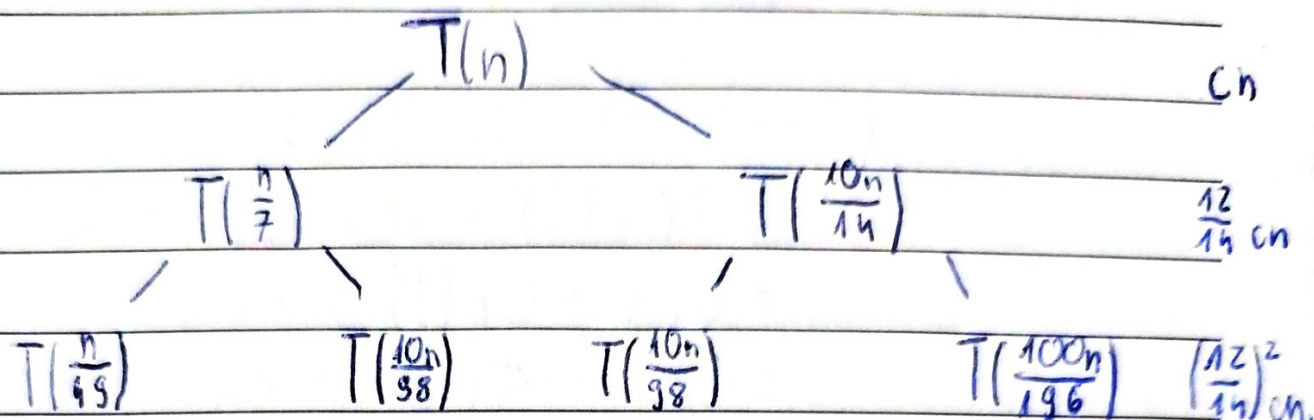
- If $l < k$, then m is smaller than the number we are looking for so we throw out the elements that are smaller and we adjust k , $M(A \setminus A_{\text{low}}, k - |A_{\text{low}}|)$

$$A_{\text{high}} = \left\lceil \frac{n}{7} \cdot \frac{1}{2} \right\rceil \cdot 4$$

$$A_{\text{low}} = \left\lceil \frac{n}{7} \cdot \frac{1}{2} \right\rceil \cdot 4$$

$$T(n) = T\left(\frac{n}{7}\right) + T\left(\frac{10n}{14}\right) + \Theta(n)$$

b)



The depth of the tree is:

- left: $\log_7 n$
 - right: $\log_{10} \frac{14}{10} n$
- unbalanced

so

$$T(n) = \sum_{i=0}^{\infty} \left(\frac{12}{14}\right)^i cn = ~~7cn~~ 7cn$$

$$T(n) \leq ~~7cn~~ 7cn$$

or

$$\begin{aligned}
 T(n) &\leq T\left(\frac{n}{7}\right) + T\left(\frac{10n}{14}\right) + cn \\
 &\stackrel{IH}{\leq} 7c\left(\frac{n}{7}\right) + 7c\left(\frac{10n}{14}\right) + cn \\
 &= cn + c\left(\frac{10n}{2}\right) + cn \\
 &= cn + 5cn + cn \\
 &= 7cn
 \end{aligned}$$

So $T(n) = O(n)$ and so M is linear in the length of A

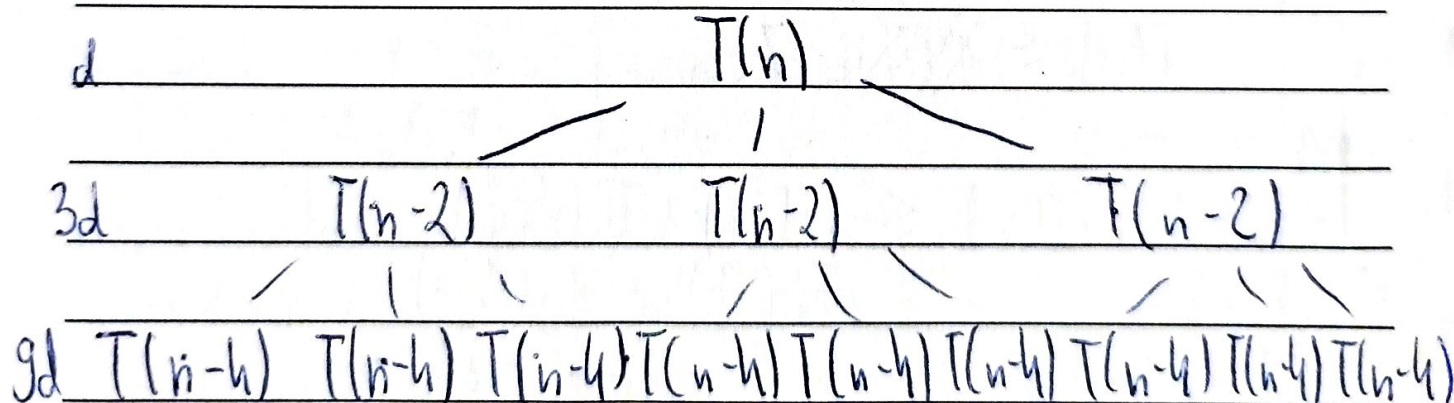
c) Let N be such that $T(n) \leq T(\frac{n}{2}) + T(\frac{10n}{14}) + cn$ for all $n > N$. No, C such that $T(n) \leq 7Ccn$ for all $n > N_0$.

$$\begin{aligned} T(n) &\leq T(\frac{n}{2}) + T(\frac{10n}{14}) + cn \\ &\stackrel{H}{\leq} 7Ccn(\frac{n}{2}) + 7Ccn(\frac{10n}{14}) + cn \\ &= Ccn + Ccn(\frac{10n}{2}) + cn \\ &= Ccn + 5Ccn + cn \\ &= 6Ccn + cn \end{aligned}$$

$$\text{where } 6Ccn + cn \leq 7Ccn \Leftrightarrow C := 1 \leq 7Ccn$$

Exercise 4.

a) $3T(n-2) + d$



- The height of the tree is $\frac{n}{2}$
- The number of leaves is $O(3^{\frac{n}{2}})$
- The cost per layer is $3^i d$

The total cost is

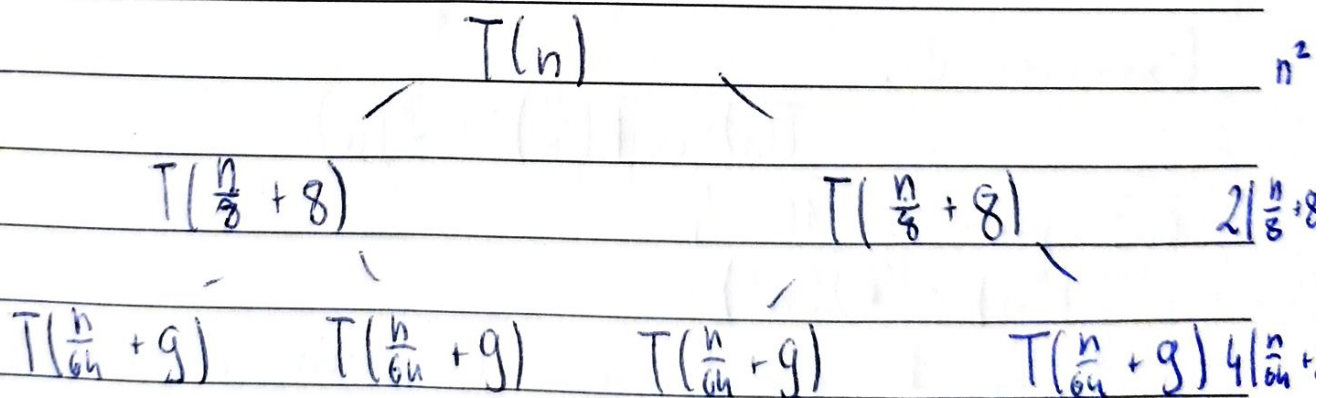
$$\sum_{i=0}^n 3^i d = d \frac{3^{\frac{n}{2}+1} - 1}{3 - 1} = d \frac{\sqrt{3}^n \cdot 3 - 1}{2} \approx d 3^{\frac{n}{2}} = d \sqrt{3}^n$$

- We claim that the complexity is
 $T(n) \in O(\sqrt{3}^n)$

The proof of $T(n) \leq C \sqrt{3}^n \cdot d$

$$\begin{aligned} T(n) &= 3T(n-2) + d \\ &\stackrel{IH}{\leq} 3C \sqrt{3}^{n-2} - 3d + d \\ &= 3C \frac{\sqrt{3}^n}{3} - 2d \\ &= C \sqrt{3}^n - 2d \\ &\leq C \sqrt{3}^n - d \end{aligned}$$

$$b) 2T\left(\frac{n}{8} + 8\right) + n^2$$



- The height of the tree is $\log_8 n$

- The cost at the bottom is $2^{\log_8 n} = n^{\log_8 2}$

The total cost is

$$\sum_{i=0}^{\log_2 n} 2^i \frac{n^2}{8^i} = n^2 \sum_{i=0}^{\log_2 n} \left(\frac{2}{8}\right)^i = n^2 \frac{\left(\frac{2}{8}\right)^{\log_2 n + 1} - 1}{\frac{2}{8} - 1} \leq 4n^2$$

We claim that the complexity is $T(n) \in O(n^2)$

The proof of $T(n) \leq Cn^2$

$$\begin{aligned} T(n) &= 2T\left(\frac{n}{8} + 8\right) + n^2 \\ &\stackrel{IH}{\leq} 2C\left(\frac{n}{8} + 8\right)^2 + n^2 \\ &= 2C\left(\frac{n^2}{64} + 2n + 64\right) + n^2 \\ &= C\frac{n^2}{32} + 4Cn + 128C + n^2 \\ &= n^2\left(\frac{1}{32}C + 1\right) + 4Cn + 128C \\ &\leq Cn^2 \end{aligned}$$

We choose C such that $\frac{1}{32}C + 1 < C$ so,
 $C > \frac{32}{31}$

Exercise 5.

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

$$\begin{aligned} a) \quad x &:= \log_2 16 = 4 \\ f(n) &= O(n^4) \end{aligned}$$

Case II of Master theorem applies

$$T(n) = \Theta(n^4 \log n)$$

$$b) \quad \gamma := \log_5 5 = 1$$

$$f(n) = \Theta(n\sqrt{n})$$

Case III applies

$$T(n) = \Theta(n\sqrt{n})$$

Check

$$5f\left(\frac{n}{5}\right) \leq c \cdot f(n)$$

$$5 \cdot \frac{n}{5} \sqrt{\frac{n}{5}} \leq c \cdot n\sqrt{n}$$

$$n\sqrt{\frac{n}{5}} \leq c \cdot n\sqrt{n}$$

$$c = \frac{9}{10}$$

$$c) \quad \gamma := \log_3 4$$

$$f(n) = \Theta(\log n)$$

Case I applies

$$T(n) = \Theta(n^{\log_3 4})$$

$$d) \quad \gamma := \log_2 16 \log_4 8$$

$$f(n) = \Theta(n\sqrt[3]{n})$$

Case I applies

$$T(n) = \Theta(n^{\log_2 16 \log_4 8})$$